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Ex. No : 09

Persistence Layer – SharedPreferences Aim

To understand the Persistence Layer in Android and implement data storage using SharedPreferences.

Procedure

1. Open Android Studio.
2. Create a new project (Empty Activity, Kotlin).
3. Design UI using:
 - EditText (for input) ○
 - Button (to save data) ○
 - TextView (to display stored data)
4. Use SharedPreferences to store data as key–value pairs.
5. Retrieve stored data and display it.
6. Run the app and verify persistence after restarting.

Code activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etName"
        android:hint="Enter Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSave"
        android:text="Save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/tvResult"
        android:text="Stored Data:"
        android:textSize="18sp"
        android:paddingTop="20dp"
```

```
android:layout_width="match_parent"
android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.persistencedemo
```

```
import android.content.Context
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?)
    { super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val etName = findViewById<EditText>(R.id.etName)
        val btnSave = findViewById<Button>(R.id.btnSave)
        val tvResult = findViewById<TextView>(R.id.tvResult)

        val sharedPref = getSharedPreferences("MyPrefs", Context.MODE_PRIVATE)

        // Load saved data
        val savedName = sharedPref.getString("name", "No data")
        tvResult.text = "Stored Data: $savedName"

        btnSave.setOnClickListener {
            val name = etName.text.toString()

            val editor = sharedPref.edit()
            editor.putString("name", name)
            editor.apply()

            tvResult.text = "Stored Data: $name"
        }
    }
}
```

Result

The application successfully:

- Stores user input using SharedPreferences
- Retrieves stored data after app restart
- Displays persistent data on screen

